Lightweight Tuberculosis Screening Pipeline for Community-based Active Case Finding

Irish Danielle Morales, Raphael B. Alampay, Ruben A. Saulog and Patricia Angela R. Abu
Department of Information Systems and Computer Science, Ateneo de Manila University, Quezon City, Philippines

***Abstract*—Tuberculosis (TB) screening increasingly relies on computer-aided detection (CAD) systems, which are accurate but output a probability score and a visual overlay designed for radiologist reading. Community-based active case finding (ACF) screens at-risk populations at remote sites that often lack a radiologist, a GPU, and reliable connectivity, leaving a non-radiologist to act on outputs meant for an expert. We present a TB screening pipeline for adult community ACF whose contribution is the integrated system, not any single model: it runs end-to-end on a commodity CPU without a GPU or a network connection, and it replaces the radiologist-oriented score and overlay with a faithful natural-language report that a non-radiologist can act on. Among ten classifiers we evaluate, FlipR is a parameter-efficient symmetry-gated network that matches the strongest baselines and transfers best to unseen acquisition sites. On the held-out test set it exceeds the World Health Organization (WHO) target sensitivity and specificity, though at an argmax operating point on a curated benchmark not directly comparable to the WHO 2025 product evaluation. For localization we benchmark lightweight detectors for both active and latent TB, the latter underdetected in prior work, and recommend a compact off-the-shelf detector that reaches near-best localization accuracy at far smaller size. The report generator is faithful by construction, asserting only what the models detected and following radiologic report convention. The deployed configuration completes a study in about 370 ms on one CPU thread without a GPU, and compute is not the cost driver for TB CAD: carrying no per-study license, the pipeline adds effectively zero marginal cost, a structural advantage over the licensing and imaging-hardware costs that dominate commercial systems. Limited TB-specific samples in public datasets constrained both design and evaluation, and our evaluation rests on a single curated public dataset that is easier than the multi-source settings ACF targets. Further work remains to characterize latency on constrained field hardware, detect specific radiographic findings, and assess the reports through a radiologist reader study.**

***Keywords*—Tuberculosis, Computer-Aided Detection, Chest X-ray, Medical Imaging, Community-based Active Case Finding**

I. Introduction

Tuberculosis (TB) causes more deaths than any other infectious disease worldwide [1]. It is caused by airborne *Mycobacterium tuberculosis* and primarily affects the lungs, presenting as pulmonary TB [3]. Pulmonary TB is typically diagnosed by assessing symptom history, chest radiography, and bacteriological confirmation. A clinician, typically a general practitioner or pulmonologist, interviews the patient for symptoms and may request a chest X-ray (CXR), which a radiologist interprets for abnormalities indicative of TB and reports in a radiographic report. A patient whose symptoms or report indicate TB has presumptive TB. Confirmation is done through smear microscopy, culture, or rapid diagnostics such as Xpert MTB/RIF or Truenat, after which the patient has bacteriologically confirmed TB and begins treatment.

Community-based active case finding (ACF) refers to an intervention where teams screen at-risk populations in community sites or through home visits, rather than waiting for symptomatic patients to present at a clinic [2]. Figure 1 shows a representative ACF workflow, where a field team carries out screening and triage on the community site and an off-site laboratory performs confirmation.

A. Computer-aided Detection of Tuberculosis

TB computer-aided detection (CAD) systems are widely used in both clinical settings and ACF. The World Health Organization (WHO) endorsed CAD as an alternative to human readers for patients aged 15 and older, and in 2025 approved six commercial products as meeting its TB screening performance standards [7]. CAD assistance has been shown to improve radiologist performance and let non-radiologists match radiologists [9], and several reviews report CAD accuracy approaching WHO targets across populations [8], [34], [35], [14].

In ACF, WHO positions TB CAD as a screening and triage tool that selects who proceeds to confirmatory testing [6], [2], and because CAD replaces the human reader, community ACF teams commonly include no radiologist [21], [22]. Despite WHO endorsement, clinicians across countries still value CAD as a complement to, not a replacement for, expert reading [33].

B. Challenges in CAD Integration in Community-based ACF

While TB CAD systems are highly accurate, CAD workflow integration, local validation, and human factors are relatively under-studied [10], [8]. ACF, in particular, faces barriers in connectivity, compute, and interpretation:

Connectivity and compute are constrained in community sites. ACF is often conducted in remote sites where connectivity is unstable and CAD software runs on laptops [21]. Implementation studies in India and Vietnam document how these conditions shape field deployment [31], [32].

CAD overlays are designed for radiologist readers, whereas ACF teams often do not have radiologists. We review the six TB CAD products approved by WHO in 2025: qXR v4.0 [15], CAD4TB v7 [16], Lunit INSIGHT CXR v3.1 [17], DrAid v2.4.4–6 [18], Genki Edge v3.4.2 [19], and InferRead DR Chest v1.0.1.1 [20] [7]. All deliver TB screening results as a probability score with visual localization through a boxed region, outline, or heatmap. Some products additionally

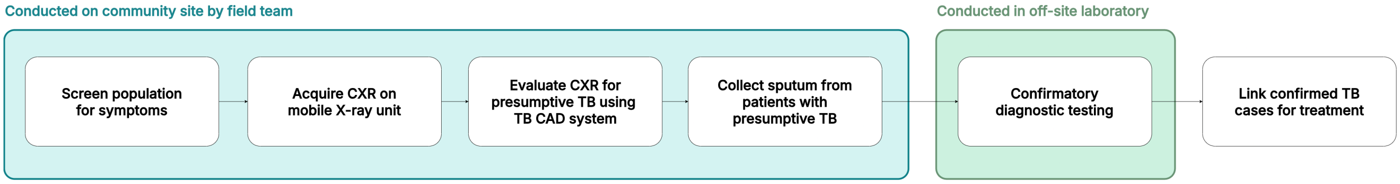

Fig. 1. A TB community active case-finding (ACF) workflow. The CAD system in ACF is often used by a radiographer rather than a radiologist.

generate broader radiology reports spanning many chest findings [18], [15], but these target radiologist reading, and the TB screening output itself remains a score and an overlay. CAD overlays also overlap poorly with expert-identified regions and omit the contextual reasoning used in diagnosis [27], and textual descriptions have been proposed as a more clinically meaningful complement to image markup [38], [36], [28], [11], [37]. Despite chest X-ray interpretation falling outside the radiographer's scope of practice, the radiographer in ACF decides whether to refer a patient for confirmatory testing using the radiograph, the patient's symptoms, and the CAD output [22], [21]. The WHO 2025 evaluation further notes that the average reader in resource-limited settings is likely less proficient than the expert radiologists these products were benchmarked against [7].

C. Related Work

TB CAD models with lightweight memory and compute have been explored in previous studies, from LightTBNet at about 1.5 million parameters [49] to lightweight hybrid networks [50], building on edge-deployed chest X-ray models for related conditions [25], [26]. LightTBNet was validated for binary TB classification on Montgomery and Shenzhen-style datasets rather than the three-class TBX11K task evaluated here, so we retrain its architecture under our protocol as the smallest classifier in our comparison. Some existing commercial CAD systems require a dedicated CAD box or separate CAD laptop to support offline screening due to compute constraints [21].

Related literature has also explored models that generate natural-language descriptions of findings. In a research prototype, Siemens Healthineers paired a combined classification-localization model [12] with a domain-adapted LLM to generate CXR findings [11]. General medical vision-language models such as LLaVA-Med [13] and CXR-LLaVA [29] demonstrate image-conditioned radiology text generation [30]. However, most work remains in general chest X-ray detection, with few specific to tuberculosis.

D. Contributions

This paper's central contribution is an integrated TB screening pipeline for community ACF, not any single model. The pipeline targets the field setting the preceding challenges describe: no radiologist on-site, often no GPU, and unreliable connectivity. It replaces the radiologist-oriented probability score and visual overlay with a faithful natural-language report that a non-radiologist reader can act on. We make three system-level contributions:

- We assemble an end-to-end screening pipeline that runs offline on a commodity laptop CPU, completing a study in about 370 ms without a GPU or a network connection, within the compute and connectivity limits of community ACF sites.
- We transform the system output from the radiologist-oriented score and overlay into a deterministic, faithful-by-construction natural-language report that asserts only what the models detected, so a non-radiologist reader receives an actionable result.
- We show through a cost analysis that compute is not the cost driver for TB CAD: the pipeline carries no per-study license and adds effectively zero marginal cost per study, a structural advantage over the licensing and imaging-hardware costs of commercial systems.

The models inside the pipeline are selected under these field constraints rather than for benchmark accuracy. Among ten lightweight classifiers we evaluate, FlipR is a parameter-efficient symmetry-gated network that matches the strongest baselines and transfers best to unseen acquisition sites. Among lightweight detectors for active and latent TB localization, the latter underdetected in prior work, we report the honest negative result that a larger custom backbone does not surpass a compact off-the-shelf detector.

We evaluate the pipeline on a single public dataset for adult screening and establish on-device feasibility on a commodity CPU, and we identify a field-population reader study and field-hardware characterization as the necessary next steps.

II. System Overview

This paper presents a lightweight TB screening pipeline that describes model findings in natural language, as shown in Figure 2. A CXR (DICOM or standard image format) is ingested from the screening unit's X-ray or uploaded directly, then scored by the TB classification module. For images flagged as TB, the object detection module identifies active and latent TB regions. The classification and detection outputs are passed to the report generation module, which produces a natural-language report of the findings based on radiologic report convention. All three stages are lightweight, for screening where no radiologist and often no GPU is present.

III. Tuberculosis Classification

The TB classification module flags each CXR as either healthy, sick non-TB, or TB as shown in Figure 2.

A. Dataset

Classification and object detection models are trained on a simplified redistribution[1] of TBX11K [39], a public CXR

[1]https://www.kaggle.com/datasets/vbookshelf/tbx11k-simplified

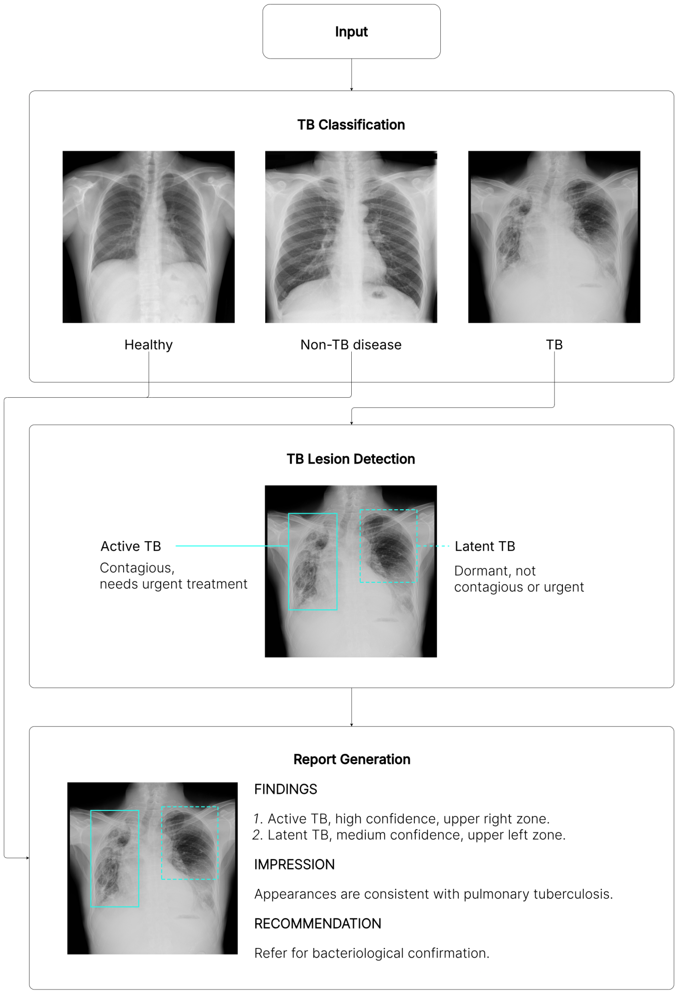

Fig. 2. Overview of the TB CAD pipeline: image classification, TB lesion detection, and report generation.

dataset with image-level labels (*healthy*, *sick non-TB*, *TB*) and bounding-box-level labels (*active-TB*, *latent-TB*).

TBX11K is selected because it is the largest public TB CXR dataset with a TB-specific sample large enough to train from scratch, expert bounding-box annotations, a sick non-TB class, and a demographic and class composition that reflect real-world clinical settings [40]. Its image-level labels are established by diagnostic microbiology and reviewed by experienced radiologists [40], the same microbiological reference standard used in the WHO 2025 product evaluation [7], rather than by image reading alone. TBX11K notes per-image patient age and sex, with most TB cases aged 20 to 60 and a male majority (72% of TB X-rays), which is representative of the general population and consistent with real-world clinical scenarios [40].

The official TBX11K test set is withheld for an online challenge, so we use only the public trainval images. Our results are therefore not directly comparable to SymFormer [40] or other challenge entries on the private test set.

Classification uses the image-level labels shown in Table I. The 8,400 public CXRs (3,800 healthy, 3,800 sick non-TB, 800 TB) are partitioned per class into train, validation, and test folds in a 70%/15%/15% ratio. All reported classification metrics are computed on the held-out test fold of 1,260 CXRs (570 healthy, 570 sick non-TB, 120 TB).

TABLE I. TBX11K DATASET COMPOSITION (IMAGE-LEVEL LABELS) FOR CLASSIFICATION

Class	Train	Val	Test	Total
Healthy	2,660	570	570	3,800
Sick non-TB	2,660	570	570	3,800
TB	560	120	120	800
Total	5,880	1,260	1,260	8,400

Per-class 70/15/15 train/validation/test split.

B. Data Preprocessing

Classification models are trained without preprocessing or data augmentation. TBX11K CXRs are acquired under a standardized posteroanterior protocol and are largely uniform in pose, so aggressive augmentation risks introducing distribution shift.

C. Model Architecture

We evaluate ten classifiers: six baselines (EfficientNet-B0 [44], MobileNetV3-Large [43], DenseNet-121 [42], ResNet-18, ResNet-50 [41], ConvNeXt-Tiny [45]), three custom models (DraxNet, Drax-MobileNetV3-Large, FlipR), and LightTBNet [49], a published TB-specific lightweight network retrained under our protocol. Full architecture of custom models are listed in Appendix A.

1) DraxNet: DraxNet (16.5M parameters) modifies a ResNet-18 backbone so the fourth stage consists of two residual blocks, each with a custom mixer (*DraxBlock*) inserted between its two convolutions. All other components are unchanged. *DraxBlock* consists of a ConvNeXt local convolutional branch followed by an optional self-attention branch, fused via stochastic-depth residuals. DraxNet, short for Dynamic Residual Attention eXchange, reserves the heavier mixer for the final, lowest-resolution stage, adding attention capacity with fewer parameters than a full attention backbone.

2) Drax-MobileNetV3-Large: Drax-MobileNetV3-Large (4.8M parameters) applies the DraxBlock to a parameter-efficient backbone. The MobileNetV3-Large feature extractor and head are unchanged, and a bottlenecked *DraxBlock* is inserted after the final feature stage. This design tests whether the DraxBlock generalizes beyond a ResNet-style backbone.

3) FlipR: FlipR (2.8M parameters) inserts a *FlipR* block between the second and third stages of a ResNet-18 truncated to three stages, with the parameter-heavy fourth stage removed. The *FlipR* block computes the difference between each feature map and its horizontally flipped counterpart, smoothed by average pooling to account for imperfect anatomical symmetry, then broadcasts a gated mask across all channels. The block is inspired by SymAttention [40], which observes that bilateral asymmetry in CXRs is a key indicator of TB. FlipR tests whether a lightweight symmetry gate can match heavier attention approaches.

D. Model Training

All models are trained from scratch with a fixed random seed, batch size 16, 512×512 input, 100 epochs, Adam optimizer at a fixed learning rate of 10^{-3}, and unweighted cross-entropy loss. No hyperparameter tuning was done. The checkpoint with the lowest validation loss is retained and evaluated on the held-out test fold.

E. Results

Classification results are reported in Table II.

The smallest model on parameters is among the largest on compute, so no single model is lightest on both axes. The top five models by accuracy (LightTBNet, FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) differ by within $\pm 1\%$ on accuracy and F1. LightTBNet uses the fewest parameters (1.3M), but its high-resolution residual design costs 14.1G MACs, more than every model except the three largest (ConvNeXt-Tiny, ResNet-50, and DenseNet-121) and nearly double FlipR. FlipR is the next smallest at 2.8M (4.0 to 4.8M for the other three), consistent with the SymFormer [40] observation that bilateral asymmetry is a strong indicator of TB. Its parameter savings come from removing the parameter-heavy final stage of ResNet-18, which runs at low resolution and so contributes little compute. FlipR roughly halves LightTBNet's compute at 7.4G but still exceeds the depthwise-separable backbones (MobileNetV3-Large 1.2G, Drax-MobileNetV3-Large 1.3G, EfficientNet-B0 2.1G), which keep both parameters and compute low. Parameter count measures the storage and memory footprint, while MACs measure compute, and the two favor different models, so a model that minimizes parameters alone can still carry a heavy compute cost.

All models exceed WHO triage targets on this test set, but not under the WHO 2025 evaluation protocol. Sensitivity ranges from 92.50 to 96.67%, above the 90% sensitivity target, and specificity ranges from 97.98 to 99.82%, above the 70% specificity target of the WHO target product profile [2]. Sensitivity is more clinically important than specificity in screening, because a missed TB case risks untreated transmission while a false positive triggers only unnecessary confirmatory testing. MobileNetV3-Large achieves the highest sensitivity, missing only 3.3% of TB cases. These figures are not directly comparable to the WHO 2025 evaluation of approved products [7], for two reasons. First, the operating point differs: we report argmax sensitivity and specificity, whereas the WHO 2025 protocol fixes sensitivity at 90% and reports specificity there (with a clinical cutoff near 40%, not 70%) alongside a partial AUC over the 40 to 60% specificity range. Second, the test set differs in difficulty: although TBX11K shares the WHO microbiological reference standard, it is curated, single-source, and class-balanced, whereas the WHO libraries aggregate multiple regions and sources, and approved products reach only 26 to 56% specificity at 90% sensitivity on them. Restated under the WHO 2025 protocol on this test set, all ten models saturate these indices, with specificity at 90% sensitivity between 99.47 and 100.00% (Spec@90sens column, Table II) and a partial AUC of 1.00 over the 40 to 60% specificity band. Every model reaches 100% sensitivity at a specificity of 80% or higher, well above the 40 to 60% band at which WHO scores approved products, so the partial AUC is pinned at 1.00. This saturation reflects the easier curated test set rather than a genuine pass of the WHO bars, and reporting these metrics on a comparable multi-region library remains future work.

The high-specificity, lower-sensitivity profile is likely attributable to TBX11K rather than model architecture. All models display high specificity and lower sensitivity, consistent with results from models in related literature trained on TBX11K. The TBX11K dataset is constructed to reward high specificity because the common failure mode on other TB datasets is over-predicting TB and achieving near-zero specificity [40].

Lightweight models are preferred for TB screening on both performance and deployment grounds. The smaller models (LightTBNet, FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) match or exceed every larger model in accuracy and F1. Due to a modest TBX11K training set, a minority TB class, and no class-imbalance handling, larger networks appear to overfit the majority healthy and sick non-TB classes rather than learn discriminative TB features. TB screening also runs in resource-limited settings often without a dedicated GPU, where both the memory footprint set by parameter count and the inference latency set by MACs matter. No single model minimizes both. LightTBNet has the fewest parameters but a high MAC cost, FlipR is the most parameter-efficient model with moderate compute, and the depthwise-separable backbones have the fewest MACs, so the deployment choice depends on whether storage or compute is the tighter constraint on the target device.

F. External Validation

Zero-shot transfer to unseen sources collapses from near-perfect to weak. The results above are computed within TBX11K. To test transfer to other acquisition sites, we run each trained classifier, without retraining, on two independent public CXR sets from different countries and scanners: the Montgomery County set and the Shenzhen set [59]. Both are binary normal-versus-TB collections, so they probe the cross-source shift that community deployment faces. Scoring TB triage as binary with $P(\mathrm{TB})$ as the positive score, mean $\mathrm{AUC}_{\mathrm{TB}}$ falls from 0.998 within TBX11K to 0.564 across the two external sets, with several models at or below chance (Table III). Figure 3 shows this collapse for the deployed FlipR classifier. The saturated in-distribution figures therefore reflect a single curated source rather than transferable screening skill, and any claim of readiness is bounded to the TBX11K distribution.

The two sites fail in opposite directions, tracking a measurable appearance shift. On Montgomery the models lose sensitivity, defaulting most TB cases to normal, while on Shenzhen they lose specificity, over-flagging normal images. The two sets differ from the TBX11K training images in opposite ways in mean pixel intensity (Montgomery darker at 113, Shenzhen brighter at 157, against 123 for TBX11K), with a Jensen-Shannon divergence of intensity histograms from the training set of 0.30 and 0.29 respectively (Figure 6). Because the failure direction follows this shift, the collapse is a covariate shift in image appearance rather than a labeling artifact.

TABLE II. TUBERCULOSIS CLASSIFICATION RESULTS

Model	AUC_{TB}	Classification metrics (%)							Computational metrics	
		Acc	Sens	Spec	Spec@90sens	Prec	Rec	F1	Params	MACs
LightTBNet [49]	0.9994	98.89	94.17	**99.82**	99.82	98.72	97.65	98.17	**1.3M**	14.1G
FlipR	0.9987	98.81	94.17	99.56	99.82	98.01	97.59	97.79	2.8M	7.4G
EfficientNet-B0 [44]	**0.9995**	98.89	95.00	99.65	**100.00**	98.29	97.87	98.07	4.0M	2.1G
MobileNetV3-Large [43]	**0.9995**	**99.21**	**96.67**	**99.82**	**100.00**	**98.97**	**98.54**	**98.75**	4.2M	**1.2G**
Drax-MobileNetV3-Large	0.9989	98.81	95.00	99.65	**100.00**	98.23	97.81	98.02	4.8M	1.3G
DenseNet-121 [42]	0.9975	98.57	92.50	99.56	99.82	97.81	96.97	97.38	7.0M	15.0G
ResNet-18 [41]	0.9982	98.17	93.33	99.21	99.91	96.71	96.90	96.80	11.2M	9.5G
DraxNet	0.9988	98.41	95.00	99.47	**100.00**	97.51	97.51	97.51	16.5M	10.9G
ResNet-50 [41]	0.9978	97.70	92.50	99.47	99.47	96.95	96.33	96.63	23.5M	21.5G
ConvNeXt-Tiny [45]	0.9952	97.14	95.00	97.98	99.82	93.63	96.58	94.97	27.8M	23.4G

AUC_{TB} is the area under the ROC curve for the TB class. Sensitivity is recall for the TB class. Specificity is recall for the non-TB class (healthy and sick non-TB combined). Precision, recall, and F1 are averaged across the three classes (healthy, sick non-TB, TB). Params is the trainable parameter count. MACs is the number of multiply-accumulate operations for one 512×512 forward pass, and FLOPs are twice this. Spec@90sens is the highest specificity at sensitivity at least 90%, the single operating point the WHO 2025 product evaluation scores [7], computed on the binary TB projection with $P(\text{TB})$ positive and healthy and sick non-TB pooled as negative. It saturates because this curated, single-source, class-balanced test set is far easier than the multi-region WHO libraries, on which approved products reach only 26 to 56% specificity at 90% sensitivity, so it is a format restatement rather than a WHO pass.

TABLE III. EXTERNAL VALIDATION: ZERO-SHOT TRANSFER AND NO-RETRAINING RECOVERY (AUC_{TB})

Model	In-dist	Montgomery			Shenzhen		
		Zero-shot	AdaBN	Crop	Zero-shot	AdaBN	Crop
LightTBNet [49]	0.999	0.470	**0.571**	0.432	0.394	0.375	**0.430**
FlipR	0.999	**0.669**	0.618	0.645	0.688	0.676	**0.705**
EfficientNet-B0 [44]	0.999	0.466	**0.602**	0.551	0.614	0.463	**0.717**
MobileNetV3-Large [43]	0.999	0.426	**0.589**	0.260	0.511	0.326	**0.688**
Drax-MobileNetV3-Large	0.999	0.571	**0.725**	0.384	0.475	0.434	**0.569**
DenseNet-121 [42]	0.997	0.511	0.537	**0.552**	0.495	0.545	**0.594**
ResNet-18 [41]	0.998	**0.613**	0.610	0.568	0.639	0.565	**0.727**
DraxNet	0.999	0.563	0.591	**0.599**	0.604	0.599	**0.754**
ResNet-50 [41]	0.998	**0.636**	0.625	0.635	0.637	0.612	**0.785**
ConvNeXt-Tiny [45]	0.995	**0.697**	–	0.599	0.613	–	**0.687**
Mean	0.998	0.562			0.567		

Each trained classifier is evaluated, without retraining, on the Montgomery (138 images, 58 TB) and Shenzhen (662 images, 336 TB) sets [59], scoring TB triage as binary with $P(\text{TB})$ as the positive score and healthy and sick non-TB pooled as negative. **In-dist** repeats the within-TBX11K AUC_{TB} from Table II. **Zero-shot** applies the model as trained. **AdaBN** recomputes batch-normalization statistics on the unlabeled target images and is not applicable to ConvNeXt-Tiny, which uses layer normalization rather than batch normalization. **Crop** restricts the input to the segmented lung field. Bold marks the best correction per model and site.

Two label-free, retraining-free adjustments recover much of the gap, and they are complementary. Real deployment can adapt to a site without relabeling, so we test two standard corrections that use only the unlabeled target images and leave all learned weights fixed. Adaptive batch normalization (AdaBN) recomputes each batch-normalization layer's running statistics on the target images [60]. Lung-field cropping restricts the input to the lungs using an off-the-shelf segmentation model [61], removing burned-in text, borders, and field-of-view differences that are known cross-dataset shortcuts [62]. The two corrections address different shift components. AdaBN recovers Montgomery, the intensity-shifted set, raising MobileNetV3-Large from 0.426 to 0.589 and Drax-MobileNetV3-Large from 0.571 to 0.725, whereas cropping recovers Shenzhen, the framing-shifted set, raising ResNet-50 from 0.637 to 0.785 and DraxNet from 0.604 to 0.754. Neither helps both sites, consistent with each addressing a distinct cause. Across the ten models with adaptation runs, applying the better correction per site raises their mean AUC_{TB} from 0.544 to 0.643, with the strongest models reaching 0.72 to 0.79. Renormalizing the score over the healthy and TB classes alone, the correct binary projection of the three-class head, adds a further 0.03 on average. These figures remain below the in-distribution values and below the WHO operating bars, so they do not establish readiness. They show instead that most of the apparent collapse is a correctable site-calibration

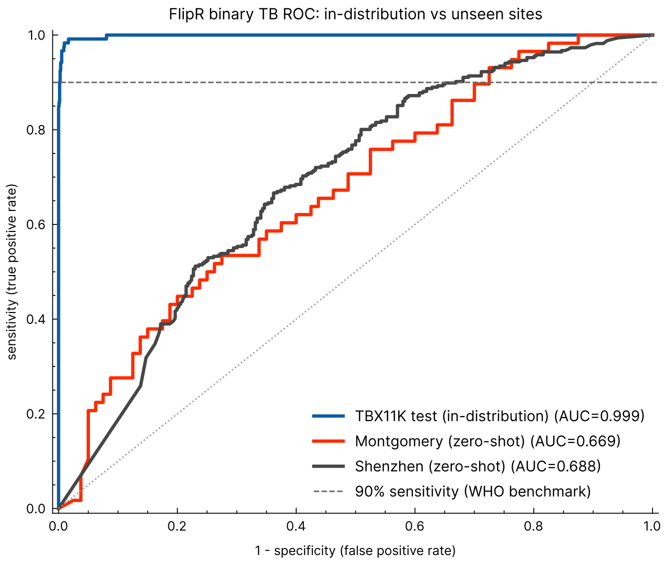

Fig. 3. Binary TB ROC for the deployed FlipR classifier on the in-distribution TBX11K test set against the two unseen sites, Montgomery and Shenzhen. The in-distribution curve saturates in the top-left corner while both external curves fall toward the diagonal, the visual form of the AUC collapse reported in Table III. The dashed line marks the 90% sensitivity operating point.

effect rather than a loss of learned TB features.

IV. Tuberculosis Lesion Detection

We frame TB lesion detection as an object detection task: the module localizes each TB region on the radiograph as a bounding box and classifies it, supplying the region-level evidence that the report generation module later renders as text.

A. Dataset

Object detection uses the TBX11K bounding-box annotations, with each region labeled active TB or latent TB. Dataset composition is shown in Table V.

Active TB denotes ongoing, transmissible infection requiring treatment. Latent TB denotes prior, inactive infection that is not contagious but may reactivate. As Figure 2 shows, active TB is more visually salient and latent TB is more subtle. A CXR with an image-level label of TB may contain bounding box labels of only active-TB lesions (630 images), only latent-TB lesions (139 images), or both (30 images). This composition is consistent with real-world conditions [39].

Differentiation between active and latent TB is relatively understudied in TB CAD. In routine reporting, a radiologist may not always state whether a lesion is suspected as active or latent TB explicitly, instead inferring it from radiographic pattern, with consolidation and cavitation suggesting active disease and fibrotic linear opacities suggesting healed disease, alongside clinical and laboratory correlation [4]. Current literature understudies whether existing commercial CAD systems can differentiate latent from active TB [21].

B. Model Architecture

We evaluate four lightweight detectors at 256px input: base YOLO26-n [46], two standard nano baselines YOLOv8n [47] and YOLO11n [48], and a custom variant in which YOLO26's default feature extractor is replaced by the DraxNet backbone used in classification. All detectors are trained from random initialization under the same detection head and training protocol. The DraxNet variant retains YOLO26's head, which isolates the backbone's effect relative to base YOLO26.

C. Model Training

Object detection follows the classification setup in training all detectors from random initialization for 100 epochs at batch size 16, but uses the Ultralytics default detection optimizer (auto-selected, base learning rate 10^{-2}) rather than the Adam optimizer at 10^{-3} used for classification. As in classification, the best checkpoint (selected by mAP) is retained and used for evaluation.

D. Results

Detection results are reported in Table IV. The WHO 2025 operating-point protocol scores a binary triage probability and has no localization analogue, so detection is evaluated with mean average precision (mAP), the standard localization metric, rather than the WHO indices reported for the classifier in Table II.

Active TB is more likely to be detected than latent TB. All four detectors localize active TB several times more accurately than latent TB, roughly three to six times at mAP_{50} and three to nine times at $\mathrm{mAP}_{50\text{-}95}$. This gap reflects the TBX11K class imbalance (active-TB regions outnumber latent-TB regions roughly 4:1) and the subtle appearance of latent-TB lesions. The TBX11K reference reports the same pattern across all detection methods, attributing it to imbalance [40], and it is consistent with human readers distinguishing latent TB less reliably than active TB [5]. The bias toward active TB also aligns with clinical triage priorities, because active TB requires urgent treatment.

The custom backbone leads active-TB localization, but by too small a margin to justify its cost. At matched input resolution and seed, the DraxNet backbone attains the best active-TB localization at 0.5865 mAP_{50} and 0.2748 $\mathrm{mAP}_{50\text{-}95}$, ahead of the stock YOLOv8n (0.5563 and 0.2582) and YOLO11n (0.5434 and 0.2446). The gain comes from replacing base YOLO26's feature extractor: DraxNet raises active-TB mAP_{50} from 0.4511 to 0.5865 ($+0.1354$) and $\mathrm{mAP}_{50\text{-}95}$ from 0.2110 to 0.2748 ($+0.0638$) over its own base, driven almost entirely by precision ($0.74 \rightarrow 0.82$) at unchanged recall (roughly 0.26). The improvement does not extend to latent TB, where DraxNet is the weakest of the four (0.1017 mAP_{50} against 0.1175 for YOLOv8n). The active-TB edge over YOLOv8n is small, about 0.03 mAP_{50}, while its cost is large: the DraxNet backbone raises the detector to 19.7M parameters against YOLOv8n's 3.2M, far above the lightweight budget this pipeline targets for edge deployment. The marginal active-TB gain does not justify roughly six times the parameters and compute, so we adopt the stock YOLOv8n

TABLE IV. TUBERCULOSIS LESION DETECTION RESULTS

Model	Active TB		Latent TB		Params	MACs
	mAP_{50}	$\mathrm{mAP}_{50\text{-}95}$	mAP_{50}	$\mathrm{mAP}_{50\text{-}95}$		
YOLO26 [46]	0.4511	0.2110	0.1070	0.0410	2.6M	0.5G
YOLO11n [48]	0.5434	0.2446	0.1166	0.0435	2.6M	0.5G
YOLOv8n [47]	0.5563	0.2582	**0.1175**	**0.0444**	3.2M	0.7G
YOLO26 + DraxNet	**0.5865**	**0.2748**	0.1017	0.0310	19.7M	3.2G

MACs is the number of multiply-accumulate operations for one 256×256 forward pass at the detection training resolution, and FLOPs are twice this.

TABLE V. TBX11K DATASET COMPOSITION (BOUNDING BOX LABELS) FOR LESION DETECTION

Annotation	Train	Val	Total
Active-TB boxes	724	248	972
Latent-TB boxes	178	61	239
Total boxes	902	309	1,211

Counts are individual bounding boxes, not images. The 799 TB images (599 train, 200 val) contain 1,211 boxes, because one image may hold several regions.

as the detection model, the strongest of the lightweight detectors and within $0.03\ \mathrm{mAP}_{50}$ of the best at a fraction of the footprint.

Recall, not precision, is the bottleneck. Precision stays high while recall lags well behind for every detector, so the detectors localize confidently but miss many annotated TB regions. Figure 4 shows recall sitting well below precision and the latent-TB curve collapsing at low confidence. The shortfall is uniform across detectors: aggregate recall stays near 0.26 to 0.28 for all four models, so no detector recovers a substantial share of the annotated lesions.

Localization is useful to confirm (rule-in) TB, but not to rule it out. A test rules in disease when a positive result confirms it and rules out disease when a negative result excludes it. The high-precision, low-recall profile indicates that a flagged lesion is likely correct and warrants focused radiologist review, but the model cannot reliably exclude TB because most annotated lesions are missed. Localization is therefore most reliable as supporting region-level evidence for the classifier and report generation modules, rather than as a standalone detector for TB. Clinicians using the model may reliably use it for rule-in diagnosis, not for rule-out diagnosis.

V. REPORT GENERATION

The report generation module turns the detector's output into a structured radiology-style report a reader can act on without interpreting a boxed region or heatmap. Example outputs are shown in Figures 2 and 5.

The module follows the two-stage detector-to-text decomposition of Danu et al. [11]. However, because the detector here emits a closed TB vocabulary rather than open findings, we replace their proprietary detector and fine-tuned language model with the public models of this paper and a deterministic module, reserving a generative model for future work that uses specific radiographic findings. For current use, template-based report generation suffices and does not require significant additional memory or compute for resource-constrained settings.

The module has two stages: an adapter that converts raw detector output into a validated structured record, and a template-based report generator that renders that record as text. The second stage receives only a structured record of what the classification and object detection modules produced. This restriction prioritizes faithfulness to model findings, because image-conditioned vision-language models such as CXR-LLaVA [29] and LLaVA-Med [13], as well as related work using LLMs [11], may report findings the detector never flagged.

A. Structured Adapter

The adapter maps classification and detection output to a fixed schema with a small, closed vocabulary. The image-level field carries the predicted label, the argmax over *healthy*, *sick non-TB*, and *TB*, together with its class probabilities. Each detected region carries three fields: its type (*active TB* or *latent TB*), a confidence band (*low*, *medium*, or *high*, thresholded at 0.5 and 0.8 on the detection score), and a location. The location is assigned by mapping the bounding-box centroid to one of nine cells in a 3×3 grid over the image (upper, middle, or lower row by left, center, or right column). An image with no detections yields an empty region list. The schema is validated before generation, so the report generator always receives well-formed input drawn from this vocabulary. The record states the type and coarse location of each region but no finer radiographic finding, because the detector labels regions only as active or latent TB.

All images are posteroanterior (PA) frontal scans. Hence, the standard radiological laterality convention is used, wherein the patient's right side appears on the image's left side and vice versa. For example, a lesion in the image's upper left corner is reported as an upper right lesion.

B. Template-Based Report Generation

Template-based report generation is a structured template over the closed schema. It renders the record as a short report in three labeled sections, mirroring the findings-to-impression structure of a radiology report [57]: a numbered FINDINGS list with one TB region per line including lesion type, model confidence band, and lesion zone, an IMPRESSION stating the screening result in clinical wording, and a RECOMMENDATION

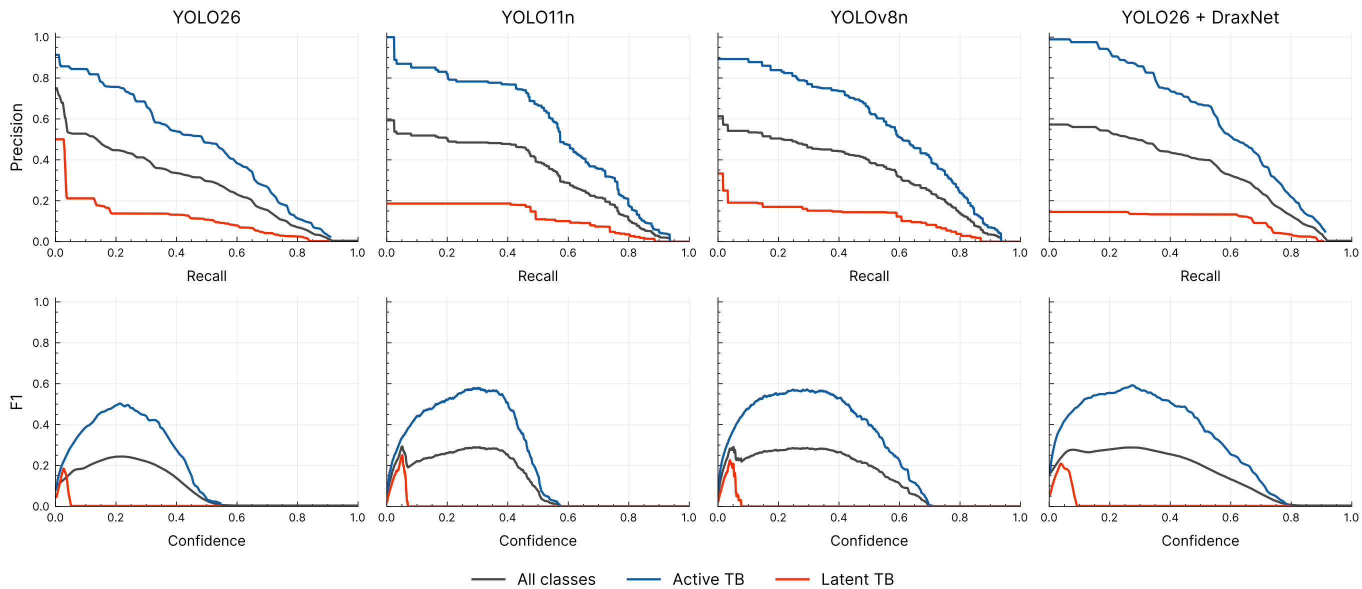

Fig. 4. Detection precision-recall (top) and F1-confidence (bottom) curves for the four detectors. The active-TB curve sits well above the latent-TB curve for every model, and the latent-TB curve collapses at low confidence, so localization functions as a high-precision rule-in cue rather than a standalone detector. The larger DraxNet variant leads on active TB, with the stock YOLOv8n close behind at a fraction of the parameters.

based on the predicted class, such as referral for bacteriological confirmation when TB is indicated.

The report describes only the detected TB regions and does not assess specific radiographic findings or cardiac, pleural, and skeletal observations. Due to lack of data, this is reserved for future work.

We chose a template over a generative model after prototyping both. An instruction-tuned language-model report generator (Phi-3.5-mini-instruct [56], run image-blind on the same record) was prototyped alongside this work, but the target is a short, fixed-format report with one line per region, which a template reproduces exactly over the closed vocabulary. The template therefore matches the generative output on faithfulness, readability, and the ordering of findings while removing any possibility of hallucination and the model inference cost, an advantage in the resource-limited settings this pipeline targets. A generative model is needed only for future work when the detector reports a richer, open set of interacting findings.

C. Faithfulness by Construction

By design, report generation is faithful by construction, defined as the generated text asserting nothing not present in the structured record. This follows the faithfulness-versus-factuality distinction drawn for abstractive summarization [51] and data-to-text evaluation [52], under which a generation is faithful if every entity it states is supported by the source, independent of whether the source is itself correct.

In previous experiments using Phi-3.5-mini-instruct for report generation, we evaluated the result for faithfulness with a deterministic verifier that parses a report for schema entities and flagging any the record does not contain: region type, location, or laterality absent from the record, a predicted label different from the record's, or an invented probability. The verifier follows an established approach in data-to-text generation, extracting the facts a generation asserts and checking them against the source records [53], and relates to the slot-error and semantic-accuracy metrics from dialogue generation that count output content not licensed by the input meaning representation [54], [55]. However, where those methods train an information-extraction or natural-language-inference model because the target text is open, the closed schema makes an exact deterministic match sufficient. Because the template renders deterministically over a validated closed schema, it cannot assert an entity absent from the record, so faithfulness holds by construction rather than as an empirical rate. The schema surface is small and closed, so the template cannot render an entity absent from the record: the two region types, three confidence bands, and nine zones, together with the empty-region case for each class, are covered by construction and checked by the report-generation tests. Since the generative approach does not provide additional benefit over the template, the generative model is dropped to avoid hallucination risk and reduce compute.

VI. On-Device Deployment

A. Feasibility

Laptops are typically used to run CAD systems in community-based ACF, resulting in limited compute. CAD is also increasingly deployed offline where network connectivity is unreliable or absent [23], [21]. This mode already operates at scale, with more than 430,000 people screened by ultraportable X-ray and CAD between 2021 and 2023 [24]. Models for community-based ACF must be able to run locally on a CPU, without a dedicated GPU or a network connection.

Our lightweight models fall at the parameter scale shown in related literature to run on this class of hardware. Lightweight chest X-ray networks have been deployed on edge devices

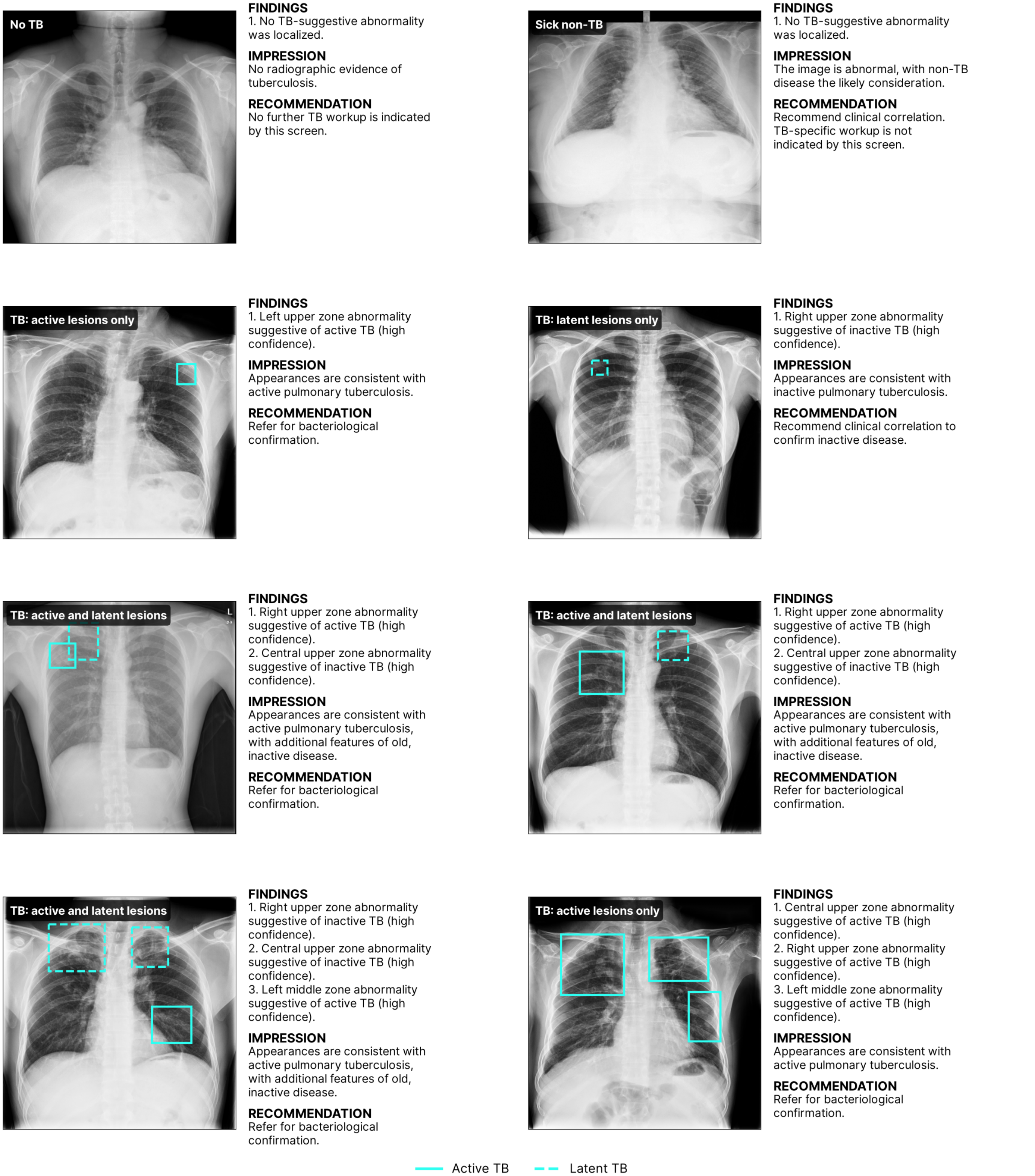

Fig. 5. Example reports on TBX11K scans, one panel per case type, produced by the deterministic report module from the detector output. The ground-truth class is overlaid at the top-left of each radiograph and the ground-truth TB regions are boxed, active TB as a solid line and latent TB as a dashed line. All panels are correct cases, so the ground truth matches the report.

and CPUs with low latency, including ResNet-18, MobileNet, and EfficientNet-B0 on an embedded board [25] and a compact classifier on edge devices [26], and for TB specifically LightTBNet reports rapid, low-memory prediction designed for

TABLE VI. ON-DEVICE CPU BENCHMARK

Model	Size (MB)	Lat. 1T (ms)	Lat. 6T (ms)	RAM (MB)
Classification				
LightTBNet [49]	**5.1**	621	214	800
FlipR	11.2	244	76	**603**
EfficientNet-B0 [44]	16.3	176	100	654
MobileNetV3-Large [43]	17.0	**89**	58	655
Drax-MobileNetV3-Large	19.4	91	**56**	631
DenseNet-121 [42]	28.4	518	259	806
ResNet-18 [41]	44.8	298	91	629
DraxNet	66.0	362	114	649
ResNet-50 [41]	94.4	682	241	819
ConvNeXt-Tiny [45]	111.4	683	217	786
Detection				
YOLO26 [46]	**5.4**	**110**	68	620
YOLO11n [48]	5.5	120	56	**608**
YOLOv8n [47]	6.2	124	**46**	610
YOLO26 + DraxNet	39.7	420	152	727

Measured on one Intel Core i7-8750H CPU under PyTorch with float32 compute, batch 1, 512×512 input, and no GPU. Lat. 1T and Lat. 6T are the median single-image forward-pass latency at 1 and 6 CPU threads over 40 runs. Size is the committed checkpoint on disk. The classification checkpoints store float32 weights, while the Ultralytics detection checkpoints store float16 weights, so detection sizes are about half of (parameters $\times$ 4 bytes). RAM is the peak resident memory of the whole inference process, which includes a fixed runtime of about 330 MB (360 MB with the detection library). These runtime figures are specific to this host and are not comparable across papers. The portable cost measures are the parameters and MACs in Tables II and IV.

embedded use [49]. Although LightTBNet carries the fewest parameters of our models (1.3M), at the 512×512 input its MAC cost (14.1G) is exceeded only by the three largest models (Table II), so a small parameter count does not by itself imply low compute. FlipR (2.8M parameters), Drax-MobileNetV3-Large (4.8M), EfficientNet-B0 (4.0M), and MobileNetV3-Large (4.2M) are at or below the size of those deployed networks, so they should run on the same devices. Their MAC cost at the 512×512 input ranges from 1.2G for MobileNetV3-Large to 7.4G for FlipR, so the depthwise-separable backbones carry the smallest compute footprint as well as a small memory footprint, while FlipR trades higher compute for a small parameter count. To test these expectations directly, we benchmarked every model on a CPU-only host. Table VI reports the single-image latency, peak memory, and on-disk size.

Every classifier completes a forward pass in under 0.7 seconds on a single thread and under 0.26 seconds on six threads, and the whole inference process peaks below 0.82 GB of memory, so the pipeline runs on a commodity laptop CPU without a GPU. The deployed configuration, FlipR followed by the YOLOv8n detector on a flagged image, runs in about 370 ms on a single thread and 120 ms on six threads. On-device latency tracks MACs rather than parameters. The depthwise-separable backbones are fastest, at about 90 ms on a single thread for MobileNetV3-Large and Drax-MobileNetV3-Large, while FlipR carries few parameters but more compute and takes 244 ms. LightTBNet makes the point sharply: despite the fewest parameters, its high-resolution design gives it the third-highest single-thread latency at 621 ms. The FlipR gap narrows to 76 against 56 ms at six threads, because FlipR's standard convolutions parallelize across cores better than depthwise convolutions. On-disk size does follow parameter count, where LightTBNet is the smallest at 5.1 MB and FlipR the next smallest. Peak memory, by contrast, tracks activation size rather than parameters: LightTBNet holds the fewest parameters yet the third-highest peak memory at 800 MB, because its high-resolution feature maps dominate the footprint. On-disk size therefore follows parameter count, while latency follows MACs and peak memory follows activation resolution.

The detectors show the same pattern. The stock nano detectors localize in 110 to 124 ms on a single thread, whereas the DraxNet-backboned variant is about four times slower at 420 ms and seven times larger on disk, which reinforces the choice of a stock lightweight detector over the DraxNet backbone. Because latency and peak memory depend on the CPU, the framework, and the thread count, these figures establish feasibility on a commodity CPU rather than on a particular field device, and they are an internally controlled comparison on one host rather than a cross-paper benchmark. A constrained single-board or ARM device would be slower, so characterizing latency on the target deployment hardware remains future work.

B. Cost

The benchmark above shows that the pipeline runs on a commodity CPU, which also shapes its cost. We report the compute cost directly, because it follows from the measured latency, and we place it against published costs for commercial TB CAD rather than asserting our own figures for products whose prices are not public.

The deployed configuration, FlipR followed by the YOLOv8n detector, completes a study in 369 ms on a single thread (Table VI), where the detector runs only on a flagged image, so this is an upper bound. A screening program of 100,000 studies per year therefore needs about 10 CPU-hours of compute per year. At a commodity cloud CPU rate near 0.045 USD per core-hour, that is under 0.50 USD per year, and on a single 300 USD offline mini-PC the same workload amortizes to about 110 USD per year while leaving the device idle for most of the day. In community active case finding, where the team already runs CAD on a laptop it owns [21], the marginal cost of an additional study is effectively the electricity, because the pipeline carries no per-study license.

Commercial TB CAD prices are negotiated and are not published as list prices [63]. The firmest public figures are the concessional bundle prices through the Stop TB Partnership Global Drug Facility, near 16,700 USD for a perpetual CAD4TB license with hardware, installation, and one year of support. A multi-product cost model for Pakistan reports per-screen costs from 0.19 to 2.73 USD, where perpetual-license products approach a low per-screen cost only at high throughput and volume-priced products stay near the 0.93 USD cost of a radiologist read [63]. Field programs report 13.31 USD per screen for CAD4TB in Zambia [64] and a 9,000 USD annual license alongside a 75,000 USD imaging device for qXR in Nigeria [65]. These figures use different denominators, base years, and cost perspectives, so we report them as context rather than as a direct comparison against the compute cost above.

Across these sources, compute is not the cost driver for TB CAD. Deployment cost is governed by licensing, imaging hardware, and program logistics. The advantage of the proposed pipeline is therefore structural rather than a lower unit price for the same service. It carries no per-study license, runs to completion on a commodity CPU without a GPU, and adds effectively zero marginal cost per study, which matters most in the decentralized, low-resource screening settings that the WHO identifies as the primary setting for automated chest X-ray interpretation [2].

VII. Discussion

A. Limitations

TB-specific samples are limited in public datasets. As of writing, TBX11K is the largest public chest X-ray dataset for tuberculosis with labeled bounding boxes, yet its TB subset remains small at 799 TB images and 1,211 TB lesion boxes. While general chest X-ray datasets exist, of these, TB-positive samples are few and insufficient to train from scratch. No public dataset contains TB-specific radiographs with reference radiographic reports, which is what a measurement of clinical correctness would require. This scarcity constrained the development and evaluation of report generation for clinical correctness.

The TB data scarcity limits how representative the evaluation population is. TBX11K is drawn from a single country, from patients presenting for hospital chest examination, and records no HIV status or prior-TB history. The WHO 2025 evaluation instead spans multiple regions, weighted toward high-burden African and South-East Asian settings, and reports lower CAD accuracy among women, people living with HIV, and those with a history of TB [7]. Reported performance here may therefore overstate what the pipeline would achieve in the higher-burden, HIV-prevalent populations that community active case finding targets. Section III-F measures one facet of this gap directly: zero-shot transfer to two other-source sets drops mean $\mathrm{AUC}_{\mathrm{TB}}$ from 0.998 to 0.564, and although label-free site adaptation recovers part of it, the corrected scores stay below the in-distribution figures. Evaluation on higher-burden, HIV-prevalent populations remains future work.

Although report generation is faithful by construction, faithfulness to the detector is not clinical correctness. A report that faithfully restates an incorrect detection is still wrong, and only reference-based or reader evaluation can report this. However, due to the lack of a TB-specific reference dataset, this is left to future work.

The system's scope is limited to screening of non-pediatric patients because the radiographic presentation of pulmonary TB is age-dependent: adult reactivation tuberculosis appears in the upper lung zones, whereas pediatric primary tuberculosis appears in the lower lung zones. Hence, detection patterns learned from a predominantly adult distribution do not transfer to children. This is consistent with the WHO endorsement of CAD as a human-reader alternative only for patients aged 15 years and older [6].

B. Future Work

Future work seeks to extend the commodity-CPU benchmark reported here (Table VI) to the target field hardware, measuring inference latency and memory on constrained single-board or ARM devices under real-world use.

Detection of full radiographic findings, such as cardiac size, pleural fluid, or non-TB abnormalities, as well as cardiac, pleural, and skeletal structures, is reserved for future work given additional data. Specific findings are more clinically meaningful for interpretation of the radiologic report, aligned with human reader interpretation of findings. This would also warrant use of a generative language model over a structured template for report generation, because synthesizing findings into a coherent report is no longer a fixed-format substitution but a genuine generation problem. This is demonstrated by recent chest X-ray systems that aggregate region-level findings into a report with an LLM [11], [58].

A reader study with radiologists and non-radiologist screeners is also recommended to evaluate the clinical correctness and coherence of the model-generated reports, particularly in comparison to human reader reports.

VIII. Conclusion

TB CAD systems are highly accurate, however challenges remain in integration for use in community ACF with limited connectivity, compute, and with no radiologist present. This paper presented a lightweight TB screening pipeline for that setting. The contribution is the integrated system, not any single model: it pairs lightweight classification and detection with deterministic, faithful-by-construction natural-language report generation and runs offline on a commodity CPU, so a non-radiologist reader receives an actionable report rather than a score and an overlay. Among the classifiers, FlipR achieves similar accuracy and F1 to the strongest baselines with a parameter-efficient design and the best zero-shot transfer to unseen sites. For localization, a comparison of lightweight detectors covers both active and latent TB, the latter underdetected in prior work, and finds that the larger custom DraxNet backbone does not justify its cost, so the compact YOLOv8n detector is preferred for deployment, within a small margin of the best accuracy at a fraction of the footprint. Across all detectors, localization functions as a high-precision rule-in cue rather than a standalone detector. Report generation provides findings in natural language for interpretation of ACF field teams and radiographers for patient referral for presumptive TB. On a commodity laptop CPU, the deployed FlipR-and-YOLOv8n configuration completes a study in about 370 ms without a GPU, and a cost analysis shows that compute is not the cost driver for TB CAD: the pipeline carries no per-study license and adds effectively zero marginal cost per study, so its advantage over commercial systems is structural rather than a lower unit price. TB-specific samples were found to be highly limited in public datasets, which constrained the design and evaluation of components. Evaluation rests on a single curated public dataset and on commodity-CPU feasibility, so a full evaluation of deployment feasibility on field hardware, extended radiographic findings, and a reader study with radiologists and radiographers in the target populations is left to future work.

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DECLARATION ON GENERATIVE AI

Generative AI tools were used to assist in writing the manuscript and source code for the study. All generative AI outputs were reverified by the authors. The conception of the research, methodology, and conclusions are the work of the authors.

DATA AND CODE AVAILABILITY

The manuscript, data, and source code for this study is publicly available at https://github.com/alive-ateneo/tb-community-screening-pipeline.

APPENDIX A
ARCHITECTURES OF CUSTOM MODELS

TABLE VII. FLIPR ARCHITECTURE (RESNET-18 TRUNCATED TO THREE STAGES)

Stage	Output	Params
Stem (conv 7×7, BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
FlipR gate	$128 \times H/8$	129
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
Head (avg-pool, FC 256→3)	3	771
Total		**2,783,684**

TABLE VIII. DRAXNET ARCHITECTURE (RESNET-18, DRAXBLOCK IN STAGE 4)

Stage	Output	Params
Stem (conv 7×7, BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
layer4 (2 DraxBlocks, 512)	$512 \times H/32$	13,699,072
Head (avg-pool, FC 512→3)	3	1,539
Total		**16,483,395**

TABLE IX. DRAX-MOBILENETV3-LARGE ARCHITECTURE

Stage	Output	Params
MobileNetV3-Large features	$960 \times H/32$	2,971,952
DraxBlock adapter (bottleneck 160)	$960 \times H/32$	575,200
Head (avg-pool, classifier 960→1280→3)	3	1,233,923
Total		**4,781,075**

APPENDIX B
DOMAIN GAP TO THE EXTERNAL SITES

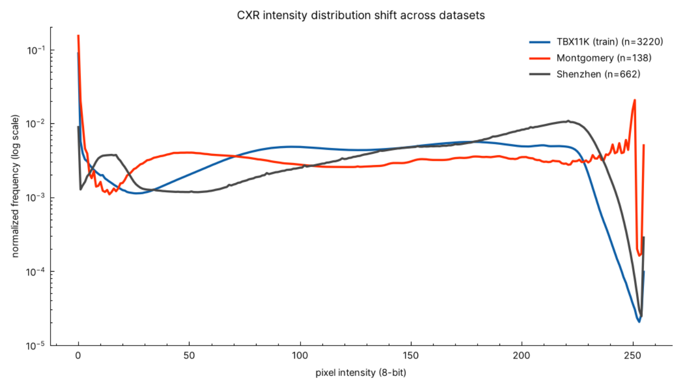

Fig. 6. Pixel-intensity distributions of the TBX11K training images and the two external sites. Montgomery is darker and Shenzhen brighter than the training set, the covariate shift in image appearance discussed in Section III-F that tracks the direction of the zero-shot failure on each site.

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